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BOARD DIPLOMA EXAMINATION, (C-16)

MARCH/APRIL—2025

DCE – FIFTH SEMESTER EXAMINATION

ADVANCED CIVIL ENGINEERING TECHNOLOGIES

Time : 3 Hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :** (1) Answer **all** questions.
(2) Each question carries **three** marks.
(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. List any three applications of IoT in security.
2. State any three applications of IoT in smart building.
3. State any three modern surveying instruments.
4. List any six losses of pre-stress.
5. List any three applications of Geomats.
6. State any three types of retaining walls.
7. State any three uses of pre-fabrication.
8. What are the precast concrete structural elements?
9. State any three characteristics of earthquake.
10. List any three solar energy utilities.

PART—B

10×5=50

- Instructions :** (1) Answer *any five* questions.
(2) Each question carries **ten** marks.
(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

- 11.** Explain the applications of IoT in (a) smart health and (b) smart transportation. 5+5=10
- 12.** Explain various segments of GPS. 10
- 13.** Explain the fundamental principles of prestressed concrete. 10
- 14.** What is post-tensioning system? Explain in detail. 3+7=10
- 15.** Explain (a) geogrids and (b) geomats. 5+5=10
- 16.** What are the different types of precast systems? Explain them. 10
- 17.** Explain the seismic resistant construction. 10
- 18.** Explain rooftop solar power generation systems. 10

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